Classification of Cancer Outcome Using Genetic and Clinical Data

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ENSIIE | Regularized Regression Method

December 12, 2025

Overview

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Project Context & Clinical Priority

Clinical Context: Breast cancer prognosis remains a major public health challenge. Accurately predicting patient survival (*Alive vs. Dead*) is critical for personalized treatment planning.

Primary Objective:

- Identify significant genetic (GeneX) and clinical biomarkers.
- Develop a robust classification model to predict `vital_status`.

The Clinical Imperative

In oncology, a **False Negative** (predicting "Alive" when patient is high-risk) is fatal.

- **Priority:** Maximize Recall (Sensitivity).
- **Method:** Threshold optimization via Youden's J.

Data Overview & Integrity (1/2)

The Datasets:

- **GeneX:** 1231×5000 (Log2 expression).
  - Challenge: High dimensionality requires feature selection.
- **Clinical:** 1231×24.
  - Contains key demographic and tumor variables.

Data Integrity Steps:

1. **Cleaning:** Removal of constant variables and redundant features (e.g., patient ID).
2. **Encoding:** Categorical features converted to dummy variables (One-Hot Encoding) for regression compatibility.

Preprocessing Strategy (2/2)

Handling Missing Data:

Two parallel workflows were tested to ensure robustness:

1. **Scenario A (Complete Case):** Row deletion.
   - Risk: High variance and loss of statistical power.
2. **Scenario B (Imputation):** Preserving $N = 1231$ samples.
   - Strategy: Mean, Mode or Median imputation for clinical data.

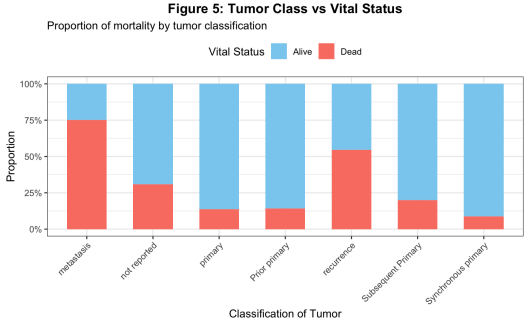
Stratified Splitting

We used a **70/30 Split stratified by** `vital_status`.

- Ensures the 1:5 class imbalance ratio is preserved in both Train and Test sets.

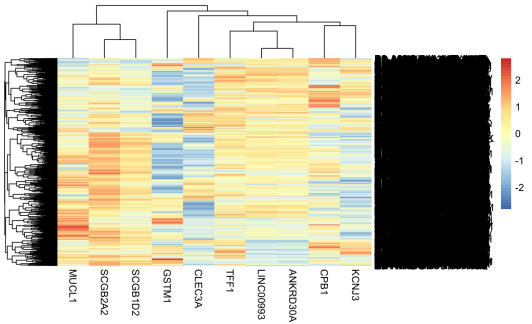
Exploratory Analysis: The Signal in the Noise

Clinical Predictors: Variables like *Tumor Classification* (Metastatic/Recurrent) show strong correlation with mortality.



[Figure: Tumor Correlation]

Genomic Complexity: 5000 genes imply high multicollinearity. *Implication:* Standard Logistic Regression will fail. Regularization (Lasso/Ridge) is mandatory.



[Figure: Genomic Heatmap]

Comparative Modeling Strategy

We evaluated candidate models under both data scenarios (Imputed vs. Dropped).

1. **Variable Selection & Regularization:**
   - **Forward Selection:** Iterative greedy approach.
   - **Lasso (L1):** Enforces sparsity. Ideal for selecting top genes from 5000.
   - **Ridge (L2):** Handles multicollinearity. Good for clinical data.
   - **Elastic Net:** Best of both worlds.

2. **Dealing with Class Imbalance:**
   - **Threshold Tuning:** We do *not* use the default 0.5 cutoff.
   - **Youden's J Statistic:** $J = \text{Sensitivity} + \text{Specificity} - 1$.
   - We select the threshold that maximizes J on the ROC curve.

Model Performance

Scenario A resulted in significant data loss ($N < 1231$) and higher variance.

Confusion Matrix (Reduced N):

$$\begin{bmatrix} 226 & 10 \\ \mathbf{43} & \mathbf{26} \end{bmatrix}$$

- **False Negatives:** 43 (High risk)
- **Precision:** 72.2%
- **Recall:** 37.7% (Poor)

Selected Features (Scenario A):

Feature
Sites of Involvement
Primary Diagnosis
AJCC Pathologic T
Tissue / Organ of Origin
Follow-ups Disease Response

Table: Top features selected in Scenario A

Performance Analysis: Scenario B

Scenario A (Row Deletion) proved unstable with high variance. **Scenario B (Imputation)** yielded robust results.

Model (Imputed Data)	Recall	Specificity	Precision	F1-Score
Lasso (GeneX Only)	0.45	0.81	0.31	0.37
Ridge (Clinical Only)	0.72	0.90	0.57	0.64
Lasso (Clinical + GeneX)	0.85	0.83	0.50	0.63
Forward Selection (Clin+Gene)	**0.92**	**0.87**	**0.59**	**0.72**

Table: Performance metrics on Test Set after Threshold Optimization.

The "Best" Model: Forward Selection (Clinical + GeneX)

Combining clinical history with genomic markers provides the best safety net.

Confusion Matrix:

$$\begin{bmatrix} 269 & 39 \\ \mathbf{5} & \mathbf{55} \end{bmatrix}$$

- **False Negatives:** Only 5 (Lowest of all models)
- **Recall:** 92% (Exceptional Sensitivity)
- **NPV:** 98.2% (High confidence in "Alive" prediction)

Top Selected Features (Coefficients):

Feature	Coeff
Primary Diagnosis	11.43
Tissue / Organ of Origin	8.51
Sites of Involvement	3.48
AJCC Pathologic T	2.50
Tumor Classification	-2.49

Table: Sorted by absolute magnitude

Advanced Approach: SMOTE + XGBoost

For comparison, a non-linear approach using Synthetic Minority Over-sampling (SMOTE).

- **F1-Score:** 0.84 (Highest overall balance)
- **Recall:** 0.86 (Comparable to Lasso)
- **Precision:** 0.83 (Significantly fewer False Alarms)

Trade-off

While XGBoost offers better precision, the Linear model is more **interpretable** for clinicians, identifying exactly which genes add risk.

Final Conclusions & Future Work

1. **Data Integrity:** Imputation and Stratified Splitting were essential for model stability.
2. **Clinical Impact:** The **Forward Selection (Clinical + GeneX)** model is recommended for screening.
   - It achieves a **Recall of 0.92**, missing only 8% of high-risk cases.
   - The high NPV (98%) makes it a safe tool for ruling out risk.
3. **Thresholding:** Optimizing via Youden's J was more effective than default cutoffs for this imbalanced dataset.

Thank You.